



The University of Texas at San Antonio®
Long School of Medicine

Center for Precision Medicine

2026

2nd Annual
Advances in Human Health through Spatial Omics and AI
Entering the Multi-Ome



San Antonio, Texas
October 5-6, 2026

MONDAY, OCTOBER 05

MEETING AGENDA - ENTERING THE MULTI-OME

LOCATION: Academic Learning & Teaching Center (ALTC) Building, Room 3.304, UTSA Lozano School of Medicine, 7703 Floyd Curl Dr., San Antonio

TIME	SESSION	SPEAKER/PRESENTER
12:30PM	REGISTRATION	
1:00PM	WELCOME	Kumar Sharma, M.D.
	OPENING REMARKS	Taylor Eighmy, PhD Inaugural President, UTSA
1:15PM	IMMUNOMETABOLISM IN PEDIATRICS	Shaun W. Jackson, MD, PhD <i>Spatial Landscape of the Kidney in Lupus Nephritis</i>
1:45PM	BIG DATA AND FEDERATED AI MODELS	Amy D. Waterman, PhD <i>Improving Cardiometabolic Health and Kidney Data Quality by utilizing a Federated Model in Two Texas Healthcare Systems</i>
2:30PM	COFFEE BREAK AND POSTERS	Networking and poster viewing
3:00PM	KEYNOTE INTRODUCTION	Francisco G. Cigarroa, MD Executive Vice President, UTSA
3:10PM	KEYNOTE SYSTEMS MEDICINE AND AI	Julia Guthrie, PhD <i>From Omics to Insight: Interactive Network Medicine for Multi-Omics Discovery</i>

4:10PM LIGHTNING TALKS

Selected Presentations From

- **Brooke Chapple, BS | Medical U. South Carolina**
Proteomic Signatures Associated with Focal Segmental Glomerulosclerosis Post-Transplant Recurrence
- **Marharyta Semenikhina, PhD | Medical U. South Carolina**
Mitochondrial metabolomic signatures of sexual dimorphism in the human kidney
- **David Hernandez-Guzman, MS | UTSA**
Visualizing Metabolic Impact: How Type 2 Diabetes Alters the Architecture of Meningeal Lymphatics
- **Thomas Kowal, PhD | Keck School of Medicine of USC**
Mapping PFAS-Induced Kidney Injury in the Mouse Kidney
- **Ana V. Quintero, PhD | UTSA**
Multimodal Spatial Metabolomics of Vascular Calcification
- **William Korinek, PhD | CEO Astrocyte Pharmaceuticals**
Surviving the Valley of Death and update on our phase II clinical trial

5:15PM RECEPTION AND POSTERS

Networking Reception and Poster Viewing

TUESDAY, OCTOBER 06

MEETING AGENDA - ENTERING THE MULTI-OME

LOCATION: Holly Auditorium/Foyer, UTSA Lozano School of Medicine, 7703 Floyd Curl Dr.

TIME	SESSION	SPEAKER/PRESENTER
8:30AM	COFFEE/TEA	
9:00AM	ORGANOIDS FOR DRUG DEVELOPMENT	James J. Hickman, PhD <i>Regulatory Acceptance of Human on a Chip Systems for Efficacy and Safety Evaluation of Therapeutics</i>
9:30AM	AI-MATCH WORKSHOP AIM-AHEAD	Amina Outub, PhD & Mark Goldberg, MD , David Hernandez-Guzman, MS, & Mignon Frances Dumanjog, BS <i>Molecules to Cells to Tissues to Systems: Multi-Network Spatial Analysis</i>
10:00AM	MULTI-OMICS AND DIAGNOSTICS	Luke Norton, PhD <i>Single Nuclei Multiomic Approaches to Dissect Transcriptional Networks in Animal and Human Liver</i> Jing Yong Ye, PhD <i>Development of Photoacoustic Aptamer Switch Sensors for Deep Tissue Biosensing and Imaging</i>
10:30AM	DIGITAL PATHOLOGY, AI, AND SPATIAL OMICS	Jeffrey B. Hodgin, MD, PhD <i>Digital Pathology and Pathomics: Omics Integration in Kidney Disease</i>

11:00AM	MULTI-OMICS CLINICAL INSIGHTS	Subramaniam Pennathur, MD <i>Lipidomics and Diabetic Kidney disease: Markers to Mechanisms</i>
11:30AM	VIRTUAL REALITY AND PATHWAYS WORKSHOP & LUNCH	Julia Guthrie, PhD & Marlene Grabner, BS, Aaron Ruiz, MS, Prince Buti, PhD
1:30PM	AI IN MEDICAL EDUCATION	Ronald Rodriguez, MD, PhD <i>The Impact of AI on Medical Education: Why Pimping is Making a Comeback</i>
2:00PM	LIGHTNING TALKS & REFRESHMENTS	Selected Presentations From Saeida Saadat, PhD University of Miami <i>BNIP3 Couples Mitophagy to Extracellular Vesicle Biogenesis in Diabetic Kidney Disease</i> Cynthia M. Estrada Zuniga, BS UTSA <i>Patient-derived organoids reveal hypoxia-driven plasticity and therapeutic vulnerabilities in pheochromocytomas and paragangliomas</i> Abhignya Jagathpally, MS University of North Texas <i>Patient-Incremental Continual Learning of Spatial Tumor Microenvironments in Multiple Myeloma</i> Soumya Maity, PhD UTSA <i>Spatial Metabolomics-Guided Identification of Circulating Biomarkers of Tissue Injury in Diabetic Kidney Disease</i> Andrew Natera, BS UTSA <i>Integrated Spatial Transcriptomic and Metabolomic Profiling Reveals Coordinated Tubular Injury and Metabolic Dysfunction in C3 Glomerulonephritis</i>

3:00PM

KEYNOTE | MULTI-OMICS IN
DIABETES

Petter Bjornstad, MD

*Cells, Space, and Disease: Decoding
Diabetes and Obesity*

4:00PM

CONCLUDING REMARKS

Kumar Sharma, MD

Professor of Medicine; Chief, Division of
Nephrology; Director, Center for
Precision Medicine

Notes:

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Group Leader at the Ludwig Boltzmann Institute for Network Medicine in Vienna, Austria



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Shaun W. Jackson, MD, PhD

Associate Professor of Pediatrics, University of Washington School of Medicine

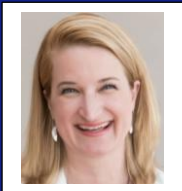


Dr. Shaun W. Jackson, MD, PhD, is an attending physician in Pediatric Nephrology and Pediatric Rheumatology at Seattle Children's Hospital and an Associate Professor of Pediatrics at the University of Washington School of Medicine, where he also directs a research laboratory focused on B cell-mediated autoimmunity. After earning his medical degree from the University of Cape Town, he completed postdoctoral training in vascular biology at Harvard Medical School, pediatric residency at Boston Children's Hospital, and dual fellowship training in pediatric nephrology and rheumatology at Seattle Children's Hospital/University of Washington. His research investigates the mechanisms by which B cells break immune tolerance and drive tissue damage in systemic lupus erythematosus and lupus nephritis, work that has earned recognition including the 2025 American College of Rheumatology Dubois Memorial Lectureship.

Amy D. Waterman, PhD

Director, Patient Engagement

Professor of Outcomes Research in Surgery, Houston Methodist Academic Institute

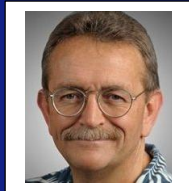


Dr. Amy D. Waterman, PhD, is a national leader in transplant health services research and the Director of Patient Engagement and Education, representing both the J.C. Walter Transplant Center at Houston Methodist Hospital and the Houston Methodist Research Institute, where she also holds the Deborah C. and Clifton B. Phillips Centennial Chair in Clinical Research in Transplant Medicine. A health services researcher and social psychologist trained at Washington University in St. Louis, with prior faculty appointments at UCLA and Washington University, she has authored more than 125 peer-reviewed publications and secured nearly \$30 million in federal grant funding for her research. Her work focuses on expanding access to kidney transplantation, reducing racial disparities, and improving patient-centered decision-making, including through Explore Transplant, an educational initiative she founded in 2009.

Notes:

James J. Hickman, PhD

Professor of NanoScience Technology Center, University of Central Florida



Dr. James "J" Hickman, Ph.D., is a Professor at the University of Nano Science Technology Center, Chemistry, Burnett School of Biomedical Sciences, and Electrical & Computer Engineering at the University of Central Florida, and co-founder and Chief Scientist of Hesperos, Inc., a pioneer in "human-on-a-chip" technology. He published the first serum-free, defined culture system for neuronal systems in 1995 and has since extended these functional in vitro systems to cardiac, muscle, hepatocyte, kidney, cancer, and epithelial cells, work that underlies Hesperos's preclinical drug discovery and toxicology platforms. A Ph.D. graduate of MIT, he holds over 165 publications, 20 book chapters, and 36 patents, and is a Fellow of the American Institute for Medical and Biological Engineering, the American Vacuum Society, the International Academy of Nanobiotechnology, and the National Academy of Inventors.

Ronald Rodriguez, MD, PhD

Professor, Department of Urology, The University of Texas at San Antonio

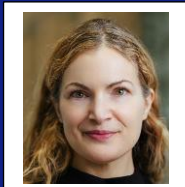


Dr. Ronald Rodriguez, MD, PhD, is Professor and the Henry B. and Edna S. Dielmann Memorial Professor of Urologic Science, and Department Chair of Urology, at UT San Antonio. A highly experienced urologic surgeon specializing in urologic oncology, he is a world expert in cryoablative technologies for kidney and recurrent prostate cancer and cares for patients with kidney, prostate, bladder, testicular, and adrenal cancers. He earned his MD/PhD through the Medical Scientist Training Program at Baylor College of Medicine, trained and served on faculty at Johns Hopkins University School of Medicine for over a decade before joining UT San Antonio in 2013, and maintains an active research laboratory investigating oxidative damage and immune regulation in kidney cancer, health disparities, and, more recently, SARS-CoV-2 treatment strategies.

Notes:

Amina Qutub, PhD

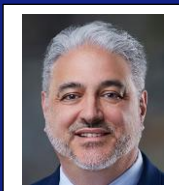
Professor of Biomedical Engineering, The University of Texas at San Antonio



Dr. Amina Ann Qutub is a Professor in Biomedical Engineering at the University of Texas. She serves as co-director of Center for Precision Medicine, assistant director of strategic partnerships and research thrust lead of the MATRIX Artificial Intelligence (AI) Consortium, and Director of the UT San Antonio Joint Graduate Group in Biomedical Engineering. Qutub is a co-founder of the AI biotech startups PaloBio, developer of behavioral-focused digital twins for athletes and individuals with atypical neurological conditions, and Selja, building AI solutions for trauma care. Her research pioneers methods at the interface of computer science, neurovascular biology, and engineering to understand how human cells communicate during growth and repair. She earned her PhD in Bioengineering from UC Berkeley and UCSF, completed postdoctoral training at Johns Hopkins University, and is a Fellow of AIMBE, an NSF CAREER awardee, and a member of the U.S. National Academies' Biotechnology & National Security Committee.

Mark P. Goldberg, MD

Professor of Neurology, The University of Texas at San Antonio

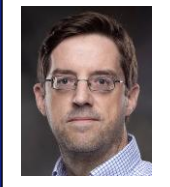


Dr. Mark P. Goldberg, MD, is Professor of Neurology and holds the Edward B. LeWinn M.D. Memorial Chair in the Department of Neurology at UT San Antonio, where he leads the multi-institutional Neurorepair Lab Group studying the biological mechanisms of neural repair and recovery after stroke and brain injury. His collaborative research includes the Glenn Biggs Institute for Alzheimer's and Neurodegenerative Diseases, the Center for Precision Medicine, and the MATRIX AI Consortium. A stroke neurologist and translational neuroscientist trained at Columbia and Stanford, he previously served as Professor and Chair of Neurology at UT Southwestern Medical Center and founded and directed the Hope Center for Neurological Disorders at Washington University School of Medicine. He is a founding member and past chair of the Lone Star Stroke Research Consortium and currently chairs the American Heart Association's national Committee on Stroke Rehabilitation and Recovery.

Notes:

Luke Norton, PhD

Associate Professor of Medicine, The University of Texas at San Antonio



Dr. Luke Norton, PhD, is Associate Professor in the Department of Medicine, Division of Diabetes at UT San Antonio. In addition, he serves as Director of the Center for Molecular Metabolism. His research investigates novel molecular, metabolic, and physiologic pathways in the liver that drive type 2 diabetes and related comorbidities such as obesity and non-alcoholic fatty liver disease (NAFLD), with his laboratory currently focused on the role of mitochondria in hepatic insulin resistance and how hepatocyte metabolic heterogeneity regulates lipid metabolism, inflammation, and fibrosis in NAFLD and Nonalcoholic Steatohepatitis. He earned his PhD in Biomedical Science from the University of Nottingham and his research has been supported by NIH funding, including an NIH/NIA K01 Career Development Award.

Jing Yong Ye, PhD

Professor of Biomedical Engineering, The University of Texas at San Antonio



Dr. Jing Yong Ye, PhD, is Professor of Biomedical Engineering and the Klesse Endowed Fellow in the Department of Biomedical Engineering at UT San Antonio, where he directs the Advanced Biophotonics and Nanotechnology Laboratory. His research focuses on biophotonics and nanobiotechnology, including the invention of a label-free biosensor using photonic crystal structures for applications such as cardiac biomarker and prostate cancer detection, as well as photoacoustic tomography for cancer diagnosis and brain imaging. His lab develops ultrasensitive, ultrafast laser-based technologies including label-free photonic crystal biosensor, a double-clad fiber-optic probe for deep-tissue in vivo imaging, and photoacoustic tomography platforms for cancer and cardiovascular applications. He holds 14 patents and has been supported by multiple NIH and USDA grants for work spanning endotoxin detection, cardiotoxicity drug screening, and nanoparticle-based cancer drug delivery.

Notes:

Jeffrey B. Hodgin, MD, PhD

Professor of Pathology, University of Michigan Medical School



Dr. Jeffrey B. Hodgin, MD, PhD, is a Professor of Pathology at the University of Michigan Medical School, where he directs the Electron Microscopy Laboratory and maintains an active research program on glomerular injury with a focus on podocyte biology. As a kidney pathologist with formal training in systems biology, his research integrates structural pathology with molecular data, and his laboratory specializes in computational pathology, artificial intelligence, and transcriptomic approaches such as single-cell RNA sequencing. He has contributed his expertise to major translational research initiatives, including the Kidney Precision Medicine Project, the Nephrotic Syndrome Study Network (NEPTUNE), the Accelerating Medicines Partnership for Lupus Nephritis, and the Human Heredity and Health in Africa (H3Africa) initiative. He is a center member at the AI and Digital Health Innovation.

Subramaniam Pennathur, MD

Professor of Internal Medicine, University of Michigan Medical School



Dr. Subramaniam Pennathur, MD, is Chief of the Division of Nephrology and the Norman Radin Chair of Nephrology Research at the University of Michigan, where he also directs the Michigan O'Brien Kidney Center and the Molecular Phenotyping and Metabolomics core of the Michigan Nutrition and Obesity Center, both NIH-funded national research centers. He trained extensively at Washington University in St. Louis, completing fellowships in Endocrinology, Diabetes and Metabolism (1996-1999) and Nephrology and Transplantation (1999-2001), where his training in analytical mass spectrometry under Dr. Jay Heinecke shaped a research career focused on the metabolic basis of diabetic complications, particularly diabetic kidney disease. He joined the University of Michigan faculty in 2006 and was named Division Chief a decade later, and his recent research spans lipidomics, metabolic crosstalk in chronic kidney disease, and lipid biomarkers linking kidney and cardiovascular health.

Notes:

Keynote Speaker: Petter Bjornstad, MD
Professor of Pediatrics and Medicine, University of Washington Medicine Diabetes Institute

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Professor of Pediatrics and Medicine, University of Washington Medicine Diabetes Institute



Dr. Petter Bjornstad, MD, is Professor of Pathology and Medicine in the Division of Metabolism, Endocrinology and Nutrition at the University of Washington, where he holds the Raisbeck Endowed Chair in Diabetes Research and serves as Director of the UW Medicine Diabetes Institute. He is a clinician-scientist funded by the NIH, AHA, and JDRF, whose research investigates the hormonal, metabolic, molecular, and hemodynamic mechanisms underlying kidney and heart disease in youth and adults with type 1 diabetes, type 2 diabetes, and obesity, and he serves as an investigator on major multicenter studies including TODAY, PERL, and the Kidney Precision Medicine Project (KPMP). Board-certified in Pediatrics and Pediatric Endocrinology, his lab is recognized for its integrative approach to early kidney and heart disease, using advanced techniques such as iohexol/p-aminohippurate clearance and functional MRI/PET imaging to assess renal and cardiac function.

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Abstracts:

1. BNIP3 Couples Mitophagy to Extracellular Vesicle Biogenesis in Diabetic Kidney Disease. **Saeida Saadat**, Wiktor Jarecki, Antonio Fontanella, Rachel Njeim, Judith Molina, George W. Burke, Sandra Merscher, Alessia Fornoni, Irena Audzeyenka and Alla Mitrofanova. *University of Miami, Miller School of Medicine, Miami, FL*.
2. Proteomic Signatures Associated with Focal Segmental Glomerulosclerosis Post-Transplant Recurrency. **Brooke Chapple**, Mykhailo Fedoriuk, Mariia Stefanenko, Lauren Ball, Daria V Ilatovskaya, Joshua H. Lipschutz, Milos N Budisavljevic, Marharyta Semenikhina, Oleg Palygin. *Medical University of South Carolina, Charleston, SC*.
3. AI-Powered Early Risk Prediction of Diabetic Kidney Disease in Adults with Type 2 Diabetes. **Claire Kyoung**, BA; Yun Shi, MD, PhD; David Katerndahl, MD; Carlos Roberto Jaén, MD, PhD; Yuanxiong Guo, PhD. *University of Texas San Antonio, San Antonio, TX*.
4. Mitochondrial metabolomic signatures of sexual dimorphism in the human kidney. **Marharyta Semenikhina**, Courtney J. Christopher, Alena Cherezova, Mariia Stefanenko Anastasiia Zavora, Justine Abais-Battad, David Mattson, Jennifer Sullivan, Joshua Lipschutz, Reilly Enos, Shawn Campagna, Oleg Palygin, Daria V. Ilatovskaya. *Medical University of South Carolina, Charleston, SC*.
5. CITE-seq reveals temporal and cell-type-specific transcriptional responses of immune cells to IL32. **Raghad Said**, Soledad Ochoa, Simon Coetzee, Nu Zhang, Marcela Haro, Kate Lawrenson. *University of Texas San Antonio, San Antonio, TX*.
6. Podocyte CerS6 Knockout Reduces C16 Ceramide and Attenuates Adriamycin-Induced FSGS. **Catarina Yee**, Andrea Salinas, Yitong Ding, Qingwei D. Zhao, Nagarjunachary Ragi, Daohong Zhou, Kumar Sharma, Guanshi Zhang. *University of Texas San Antonio, San Antonio, TX*.
7. How Reliably Do Large Language Models Answer Stroke Questions? A Comparative Analysis Across Models and Clinical Topics. **Mina Ghayour**, Jack Herz, Nithila Thangaraju, Naiesa Freeman, Jessica Pak, Brandon Jaafar, Jay Vason, Sepehr Seifi, Ali Seifi. *University of Texas San Antonio, San Antonio, TX*.
8. Plasma Adenine as an Early Biomarker of Mortality Following Severe Hemorrhage and Zone 1 REBOA in a Polytrauma Swine Model. **Dena Norouzi, BS**; Maria Castaneda, MS; Jae Hyek Choi, PhD; Kaysie Sachs; Katherine Boutin, BS; R. Madeline Paredes, PhD; Denise Manfrini, MD; Mark Foster, MD; Aaron Alindogan, MD; Subrata Debnath, Prince Buti, Nagarjunachary Ragi. Joseph K Maddry, MD; Allyson A. Mireles, PhD; Kumar Sharma, MD, Patrick C. Ng, MD. *University of Texas San Antonio, San Antonio, TX*.
9. A Quantitative Human Neurogenesis Model to Screen Functional Restoration. **Mariam Dayeh** and Amina Qutub. *University of Texas at San Antonio, San Antonio, TX*.
10. Podocyte CerS6 and C16 Ceramide in Aging-Related Renal Changes in Marmosets. **Andrea Salinas**, Catarina Yee, Yitong Ding, Qingwei D. Zhao, Hak Joo Lee, Daohong Zhou, Kumar Sharma, Guanshi Zhang. *University of Texas San Antonio, San Antonio, TX*.
11. Patient-derived organoids reveal hypoxia-driven plasticity and therapeutic vulnerabilities in pheochromocytomas and paragangliomas. **Cynthia M Estrada Zuniga**, Xiaojing Wang, Maite Calucho, Mehwish Noreen, Hector Gonzalez-Cantu, Samantha Mossuto, Jonathan N. Levi, Bethany Landry, Qianjin Guo, Kailee A. Rutherford, Ziming Cheng, Andreanne Sannajust, Sapna Raghunathan, Sanskriti Balaji, Viviane Nascimento da Conceicao, Summer Norris, S. Yusra Munawar, Huyen Thi-Lam Nguyen, Jacob De Rosa, Gail Tomlinson, James X. Wu, Michael W. Yeh, Masha J. Livhits, Anand Vaidya, James Powers, Ron Lechan, Art Tischler, Miao Zhang, Paul Graham, Camilo Jimenez, Mirko Peitzsch, Nicole Bechmann, Graeme Eisenhofer, Yanli Ding, Mio Kitano, Alice Soragni, Patricia L M Dahia. *University of Texas San Antonio, San Antonio, TX*.
12. Accuracy of Large Language Models in Adult Critical-Care Management. **Brandon Jaafar**, Carson Ward, Jessica Pak, Mina Ghayour, Jack Herz, Nithila Thangaraju, Jay Vasan, Sepehr Seifi, Ali Seifi MD. *University of Texas San Antonio, San Antonio, TX*.

13. LPS-induced Membrane Lipid Remodeling Confers Ferroptosis Resistance in Bone Marrow-Derived Macrophages. **Thuy Pham Xuan** and Chia George Hsu. *University of Texas at San Antonio, San Antonio, TX.*
14. Visualizing Metabolic Impact: How Type 2 Diabetes Alters the Architecture of Meningeal Lymphatics. **David Hernandez-Guzman** and Amina Ann Qutub. *University of Texas at San Antonio, San Antonio, TX.*
15. Mapping PFAS-Induced Kidney Injury in the Mouse Kidney. **Thomas Kowal**, Xinran Wang, Hongxu Wang, Kelly Street, Lu Zhang, Jesse Goodrich, Jonathan Nelson. *Keck School of Medicine of USC, Los Angeles, CA.*
16. Adenine Drives Mitochondrial and Innate Immune Stress in Type 1 Diabetes. **Falguni Das**, Jean C. Bopassa, Ana Victoria Quintero, Petter Bjornstad, Kumar Sharma. *University of Texas San Antonio, San Antonio, TX.*
17. Multimodal Spatial Metabolomics of Vascular Calcification. **Ana V. Quintero**, Falguni Das, Aleksandra B. Gruslova, Marc D. Feldman, Kumar Sharma. *University of Texas San Antonio, San Antonio, TX.*
18. Association Between Traumatic Brain Injury and Post-Traumatic Stress Disorder: Evidence from a Large Commercially Insured Population. **Sandra Villegas Ibarra** and Jeffrey T. Howard. *University of Texas San Antonio, San Antonio, TX.*
19. Integrated Spatial Transcriptomic and Metabolomic Profiling Reveals Coordinated Tubular Injury and Metabolic Dysfunction in C3 Glomerulonephritis. **Andrew Natera**, Prince A. Buti, Xiaojing Wang, Sarwat Gilani, Marcela Haro, Kate Lawrenson, Balakuntalam S. Kasinath, Kumar Sharma. *University of Texas San Antonio, San Antonio, TX.*
20. Accuracy and Reproducibility of Large Language Models on Pediatric Guideline-Concordance Questions. **Alexandra Ackerman**, Nithila Thangaraju, Megan McAfee, and Ali Seifi. *University of Texas San Antonio, San Antonio, TX.*
21. Metabolomic Signatures Associated with Renal Dysfunction in IgA Nephropathy. **Prince Buti**, Nagarjunachary Ragi, Subrata Debnath, Xiaolan Zhang, Brad H. Rovin, Kumar Sharma. *University of Texas San Antonio, San Antonio, TX.*
22. Spatial Metabolomics-Guided Identification of Circulating Biomarkers of Tissue Injury in Diabetic Kidney Disease. **Soumya Maity**, Ray Zhang, Jeffrey Hodgins, Ana C Ricardo, Anand Srivastava, James P Lash, Marlene Grabner, Guthrie Julia, Sushrut S Waikar, Insa M Schmidt, Kumar Sharma. *University of Texas San Antonio, San Antonio, TX.*
23. Patient-Incremental Continual Learning of Spatial Tumor Microenvironments in Multiple Myeloma. **Abhignya Jagathpally** and Bishnu Sarker. *University of North Texas, Denton, TX.*
24. Oral Amino Acids supplementation to normalize post-dialysis plasma essential amino acids in hemodialysis patients. **Anwita Surakanti**, Nagarjunachary Ragi, PhD; Prince A. Buti, PhD; Kartik Kovvuru, MD; Soumya Maity, PhD; Subrata Debnath, MB.BS., PhD. *Health Careers High School, San Antonio, TX.*
25. An untargeted to targeted metabolomics workflow for biomarker discovery. **Nagarjunachary Ragi**, Debnath Subrata, Maria Castañeda, Allyson Araña Mireles, Patrick C Ng, Kumar Sharma. *University of Texas San Antonio, San Antonio, TX.*
26. From Spatial Multi-Omics to Insight: A Network Medicine Workshop with Augmented Reality Visualization. **Marlene Grabner**. *University of Vienna, Vienna.*
27. Hypocaloric Diet Modifies Dynamic Metabolomic Responses Across Glycemic and Postprandial States in Obesity and Type 2 Diabetes. **Hussein Zaitoon, MD**, John Adams, Nagarjunachary Ragi, Prince Buti, Subrata Debnath, Kumar Sharma, MD, Renata Belfort De Aguiar, MD. *University of Texas San Antonio, San Antonio, TX.*

Acknowledgments – funding, sponsors & institutional support

Novartis Pharmaceuticals Corporation
The George M. O'Brien Kidney Research Centers
The National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)
The University of Washington Diabetes Institute
The Ludwig Boltzmann Institute for Network Medicine, Vienna
Texas Society of Nephrology
Joe R. and Teresa Lozano Long School of Medicine, The University of Texas at San Antonio

AHHA 2026 Organizing Committee

Kumar Sharma, MD, FAHA, FASN

Director, Center for Precision Medicine; Chief, Division of Nephrology; Professor of Medicine; Hillis Chair of Medicine, Vice Chair for Research, Department of Medicine, UT San Antonio

James D. Lechleiter, PhD

Co-Director, Center for Precision Medicine; Professor, UT San Antonio

Subrata Debnath, MBBS, PhD

Project Coordinator, Center for Precision Medicine; Assistant Professor of Medicine, UT San Antonio

Amina Qutub, PhD

Professor, Biomedical Engineering and Chemical Engineering; Burzik Professor in Engineering Design; Research Lead and Assistant Director of Strategic Partnerships, MATRIX AI Consortium, UT San Antonio

Mark P. Goldberg, MD

Professor of Neurology; Center for Precision Medicine, UT San Antonio

Jonathan A. L. Gelfond, MD, PhD

Associate Professor and Chief, Division of Biostatistics, Dept. of Population Health Sciences, UT San Antonio

Marc D. Feldman, MD

Professor of Medicine, Janey and Dolph Briscoe Division of Cardiology, UT San Antonio

Jing Yong Ye, PhD

Klesse Endowed Fellow and Professor of Biomedical Engineering, UT San Antonio

James Bynum, PhD

Professor and Vice Chair of Research, Department of Surgery; Executive Director, Trauma Research and Combat Casualty Care Collaborative (TRC4), UT San Antonio

Afaf Saliba, PhD

Assistant Professor of Medicine; Center for Precision Medicine, UT San Antonio

Thank you to the faculty, staff, and trainees of the Center for Precision Medicine for their support.

Special thanks to CME Office, particularly Ms. Cynthia Garza, for their invaluable assistance.

We also extend our sincere appreciation to the Nephrology administration, notably Raquel Reyna, for their outstanding support and coordination.

THANK YOU FOR JOINING US

Thank you for being part of **AHHA 2026: Advances in Human Health through Spatial Omics & AI**.

We are grateful to our speakers, participants, trainees, collaborators, and partners for sharing their expertise, ideas, and enthusiasm. We hope these two days inspire new conversations, collaborations, and discoveries that continue well beyond AHHA.

We look forward to staying connected and welcoming you again at future AHHA events.

For future questions or opportunities to connect with the Center for Precision Medicine, please contact nath@uthscsa.edu.

With appreciation,

The AHHA 2026 Organizing Committee